

Incident Response

Site:	IT-Training
Course:	Information Protection Services
Book:	Incident Response

Printed by:	Roman K.
Date:	Friday, 27 March 2026, 1:07 PM

Description



Incident Response

Didactic textbook · Version: 1.0

Topics: Incident Response - Preparation - Detection -
Containment - Recovery - Post-Incident

Goal: practical incident response mindset

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1. Introduction / Введение

1 Introduction to Incident Response

Despite extensive security measures, security incidents cannot be completely prevented. Systems can be compromised, attackers may exploit vulnerabilities, or misconfigurations can lead to data loss.

Incident response describes the structured process through which organizations respond to such security incidents, limit damage, and restore systems.

Definition:

Incident response is the organized process for detecting, analyzing, containing, and resolving security incidents in IT systems.

Why Incident Response Is Important

A security incident can have significant consequences for organizations. Without a structured response, attacks can spread and cause even greater damage.

Objectives of Incident Response:

- Detect security incidents quickly
- Limit damage
- Restore systems
- Prevent future incidents

Examples of Security Incidents

- Malware infection of a system
- Unauthorized access to a user account
- Data exfiltration from a database

- Denial-of-service attack

Take-away

Incident response ensures that organizations can respond to security incidents in a structured and efficient way.

2. Definition / Определение

2 What is an Incident

An incident is an event that can affect the security of an information system.

Such events often affect the fundamental security objectives of information security.

Relation to the CIA Triad

- **Confidentiality** – unauthorized access to data
- **Integrity** – manipulation of information
- **Availability** – failure of a system

Examples of Incidents

- Malware on a server
- compromised user accounts
- data theft
- failure of critical systems

Not every unusual event automatically represents an incident. Often several indicators must be analyzed before an incident is confirmed.

3. Event vs Incident vs Breach / Событие против инцидента против нарушения

3 Event vs Incident vs Breach

In information security, different terms are used to describe security events. These terms have different meanings.

Term	Description
Event	Observed activity in a system
Incident	Security-related incident
Breach	Confirmed data loss or unauthorized access

Example

- **Event:** unusual login activity
- **Incident:** compromised user account
- **Breach:** access to confidential data

Distinguishing these terms is important in order to correctly assess security incidents and respond appropriately.

4. Incident Response Plan / План реагирования на инциденты

4 Incident Response Plan (IRP)

An Incident Response Plan (IRP) describes how an organization responds to security incidents. The plan defines processes, responsibilities, and measures for handling incidents.

Without a clearly defined Incident Response Plan, security incidents may lead to delays, incorrect decisions, or unclear responsibilities.

Definition:

An Incident Response Plan is a documented strategy for the structured handling of security incidents.

Contents of an Incident Response Plan

- definition of security incidents
- incident response processes
- roles and responsibilities
- communication channels
- documentation requirements
- recovery procedures

Incident Response Team

An incident response team is responsible for executing the Incident Response Plan. Depending on the organization, this team may consist of various roles.

- **Incident Commander** – coordinates the incident
- **Security Analysts** – analyze the attack

- **System Administrators** – implement technical measures
- **Communication / Management** – internal and external communication

Why an IRP Is Important

A clearly defined Incident Response Plan enables organizations to handle security incidents more quickly and in a structured manner.

Take-away

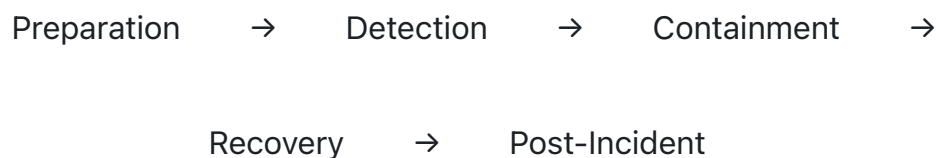
An Incident Response Plan ensures that security incidents are handled systematically and in a coordinated way.

5. Incident Response Lifecycle / Жизненный цикл реагирования на инциденты

5 Incident Response Lifecycle

Incident response typically follows a structured process known as the incident response lifecycle.

This lifecycle describes the most important phases in handling a security incident.



The Five Phases of the Incident Response Lifecycle

- **Preparation**
Preparation for potential security incidents.
- **Detection**
Identification of a security incident.
- **Containment**
Limiting damage and containing the attack.
- **Recovery**
Restoration of affected systems.
- **Post-Incident**
Analysis of the incident and improvement of security measures.

Relation to Well-Known Frameworks

The incident response lifecycle is similar to well-known security frameworks such as the NIST Incident Handling Process or the SANS Incident Response Model.

Incident Severity

Severity	Description
Low	minor security event
Medium	limited system impact
High	major system compromise
Critical	significant data breach or system outage

Take-away

The incident response lifecycle provides a structured approach to handling security incidents.

6. Preparation / Подготовка

6 Preparation

Preparation is the most important phase of the incident response lifecycle. In this phase, organizational and technical prerequisites are established so that security incidents can be detected quickly and handled effectively.

Without sufficient preparation, an incident response team cannot analyze the attack nor respond appropriately.

Basic Principle:

The better prepared an organization is, the faster security incidents can be detected and contained.

Objectives of the Preparation Phase

- define incident response processes
- establish monitoring and logging
- implement tools for incident detection
- train employees
- create documentation

Organizational Preparation

In addition to technical measures, organizational preparation also plays an important role in the incident response process.

- definition of the incident response team
- create an incident response plan
- define communication channels
- assign responsibilities

- conduct training and exercises

Technical Preparation

Technical preparation includes the implementation of security tools and monitoring systems that help detect security incidents.

- system monitoring
- security logging
- intrusion detection systems
- endpoint detection and response
- vulnerability scanning

Training and Exercises

Incident response processes should be tested regularly. Simulations and exercises help teams respond faster and in a more coordinated way during real incidents.

Take-away

Preparation ensures that organizations are technically and organizationally prepared for security incidents.

Preparation – Tool Landscape

Successful incident response depends heavily on the deployed security infrastructure.

Without appropriate tools, security incidents can neither be detected nor analyzed.

Basic Principle:

Incident detection is based on a combination of multiple security tools and monitoring systems.

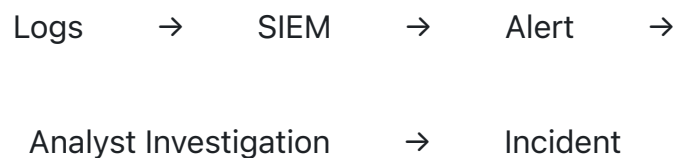
Important Security Tools

- monitoring systems
- security logging
- endpoint detection and response
- backup & restore systems
- vulnerability scanners
- patch management
- asset management
- security auditing

6.1. Detection Sources / Источники обнаружения

SOC Detection Pipeline

In Security Operations Centers (SOC), various data sources are analyzed in order to detect security incidents.



Incident Detection Sources

Security incidents can be detected from various sources.

- Endpoint Detection and Response (EDR)
- Intrusion Detection Systems (IDS)
- Security Information and Event Management (SIEM)
- [Firewall](#) Alerts
- User Reports
- Threat Intelligence

In many organizations, several of these sources are combined to enable the most comprehensive detection of security incidents possible.

Detection Stack Architecture

In modern Security Operations Centers, various data sources are combined to detect security incidents.

Network Sensors

Host Sensors

Logs

Threat Intelligence



SIEM



SOC Analysts



Incident Response

6.2. Monitoring / Мониторинг

6.1 Monitoring

Monitoring describes the continuous observation of IT systems and networks.

The goal is to detect unusual activities early that could indicate a security incident.

Definition:

Monitoring is the continuous observation of system states, network traffic, and user activities.

Why Monitoring Is Important

Many attacks remain undetected for a long time if systems are not actively monitored. Monitoring helps detect anomalies at an early stage.

Examples of Monitored Activities:

- network traffic
- system resources
- login activities
- file access
- process activities

Types of Monitoring

- system monitoring
- network monitoring
- security monitoring
- application monitoring



6.3. Logging / Ведение журнала

6.2 Logging

Logs are one of the most important sources of information when analyzing security incidents.

They document activities within systems and make it possible to trace attacks.

Definition:

Logging describes the systematic recording of events within a system.

Typical Log Sources

- Operating System Logs
- Application Logs
- [Firewall](#) Logs
- Authentication Logs
- Network Device Logs

Why Logs Are Important

Logs enable forensic analysis of security incidents.

They help understand the cause of an attack and reconstruct the attack path.

6.4. EDR

6.3 Endpoint Detection and Response (EDR)

Endpoint Detection and Response (EDR) systems monitor endpoints such as servers, workstations, or laptops for suspicious activities.

These systems collect security-relevant telemetry data and analyze it to detect attacks as early as possible.

Definition:

EDR is a security solution for detecting, analyzing, and responding to threats on endpoints.

Functions of EDR Systems

- monitoring of processes
- analysis of system activities
- detection of malware
- automatic alerting
- isolation of compromised systems

EDR in the Incident Response Process

EDR systems provide important information for the analysis of a security incident. They make it possible to trace attack activities on affected systems.

Example:

- EDR detects a suspicious process
- system is automatically isolated
- incident response team is alerted



6.5. Backup / Резервная копия

6.4 Backup & Restore

Backups play a central role in the incident response process. They make it possible to restore data and systems after a security incident.

Especially in ransomware attacks, working backups are often the only way to restore systems.

Basic Principle:

Backups enable the restoration of compromised systems after an incident.

Backup Strategies

- Full Backup
- Incremental Backup
- Differential Backup

The 3-2-1 Backup Rule

- 3 copies of the data
- 2 different storage media
- 1 offsite backup

Restore Tests

Backups are only useful if a restoration can be successfully performed.

Therefore, restore processes should be tested regularly.

6.6. Updates / Обновлять

6.5 Updates / Patch Management

Many cyberattacks exploit known vulnerabilities in software or operating systems.

Patch management ensures that such vulnerabilities are fixed through security updates.

Definition:

Patch management describes the process for identifying, evaluating, and installing software updates.

Typical Patch Management Steps

- identification of available updates
- evaluation of security relevance
- testing of updates
- deployment
- verification of installation

Patch Management in Incident Response

During an incident, patches can play an important role in closing exploited vulnerabilities.

6.7. Vulnerability Scan / Сканирование уязвимостей

6.6 Vulnerability Scanning

Vulnerability scanning describes the automated process for identifying vulnerabilities in IT systems.

These vulnerabilities can be exploited by attackers and therefore represent a potential security risk.

Definition:

Vulnerability scanning is the automated analysis of systems for known security vulnerabilities.

Objectives of Vulnerability Scanning

- identification of security vulnerabilities
- prioritization of risks
- support for patch management
- improvement of the security posture

Examples of Detected Vulnerabilities

- outdated software versions
- insecure configurations
- open network ports
- missing security updates

6.8. Assets, Docs and Auditing / Активы, документация и аудит

6.7 Asset Management

Asset management describes the administration of all IT systems and resources within an organization.

Without a complete overview of existing systems, it is difficult to correctly analyze security incidents.

Basic Principle:

You can only protect and monitor systems that are known.

Typical IT Assets

- servers
- workstations
- network devices
- cloud systems
- applications

Asset Inventory

Many organizations maintain an asset inventory that documents all systems.

This inventory significantly facilitates the analysis of security incidents.

6.8 Documentation

Comprehensive documentation is an important part of incident response preparation.

Documentation helps teams better understand systems and respond to

incidents faster.

Typical Types of Documentation

- network diagrams
- system architecture
- incident response processes
- system configurations

Well-documented systems significantly simplify the analysis of a security incident.

6.9 Auditing

Security auditing describes the regular review of security measures within an organization.

Audits help identify vulnerabilities early and improve security processes.

Types of Security Audits

- internal audits
- external audits
- compliance audits
- security assessments

Regular audits help organizations continuously improve their security posture.

6.9. SIEM

6.10 Security Information and Event Management (SIEM)

Security Information and Event Management (SIEM) systems play a central role in modern Security Operations Centers.

They collect and analyze log data from various systems in order to detect security-relevant events.

Definition:

A SIEM system collects, correlates and analyzes security events from various sources to detect potential security incidents.

Functions of a SIEM System

- collection of log data from different systems
- correlation of security events
- detection of attack patterns
- creation of security alerts
- support for incident investigations

Typical Log Sources for SIEM

- Operating Systems
- Firewalls
- IDS / IPS systems
- Endpoint Detection systems
- Cloud services
- Application logs

Event Correlation

One of the most important functions of a SIEM system is event correlation.

A single log entry is often harmless. Only the combination of several events may indicate an attack.

Example:

- multiple failed logins
- login from an unusual location
- access to sensitive files

These events together may indicate an account compromise.

SIEM in the Incident Response Process

SIEM systems enable security teams to detect security incidents faster and analyze their origin.

7. Detection / Обнаружение

7 Detection

The detection phase describes the process of identifying a potential security incident.

In this phase, security-relevant events are analyzed to determine whether an incident has occurred.

Definition:

Detection describes the process in which security-relevant events are recognized and identified as potential incidents.

Typical Detection Sources

- SIEM alerts
- EDR alerts
- IDS / IPS alerts
- [firewall](#) alerts
- user reports
- threat intelligence

False Positives

Not every security alert automatically means that an incident has occurred.

Many alerts represent so-called false positives, meaning false alarms.

Example:

- unusual login by an administrator
- legitimate software update process
- internal vulnerability scan

Detection Workflow

Alert → Investigation → Incident Confirmation

7.1. Immediate Actions / Незамедлительные действия

7.1 Immediate Actions

As soon as a security incident has been detected, initial actions must be taken. These actions are referred to as immediate actions.

The goal is to quickly assess the incident and prevent further damage.

Basic Principle:

Immediate actions serve to quickly analyze an incident and initiate initial countermeasures.

Incident Triage

Incident triage describes the initial evaluation of a security incident. During this process it is determined how critical an incident is and which actions are necessary.

Triage Questions:

- Which systems are affected?
- Which data could be affected?
- How critical is the incident?
- Does the incident response team need to be activated?

Typical Immediate Actions

- document the incident
- identify affected systems
- perform further analysis
- inform the incident response team

Alert Investigation Workflow

Alert → Triage → Investigation →

Incident Response

8. Containment / Сдерживание

8 Containment

After a security incident has been detected and confirmed, the attack must be contained as quickly as possible. This step is referred to as containment.

The goal of containment is to prevent the spread of an attack and limit further damage.

Definition:

Containment describes measures that stop an attack or limit its impact.

Objectives of Containment

- prevent further spread of the attack
- protect critical systems
- reduce damage
- preserve forensic evidence

Containment Decision

Containment measures must be carefully planned. Overly aggressive actions can destroy important evidence or take critical systems offline.

8.1. Containment Actions / Меры по сдерживанию

8.1 Short-Term Containment

Short-term containment includes measures that are implemented immediately to quickly contain an attack.

These measures are often carried out within minutes or hours after incident detection.

Typical Short-Term Measures:

- disconnect a system from the network
- disable compromised user accounts
- block IP addresses
- stop suspicious processes

Example

An endpoint detection system detects malware on a workstation.

- EDR automatically isolates the system
- network connection is disconnected
- incident response team begins analysis

8.2 Long-Term Containment

Long-term containment includes measures that aim to permanently prevent an attack from succeeding again.

These measures are often implemented after the initial stabilization of the system.

Typical Long-Term Measures:

- installation of security updates
- changing passwords
- network segmentation
- improving access controls

8.3 System Isolation and Shutdown

A common containment measure is the isolation of a compromised system.

The system is disconnected from the network to prevent further communication with attackers.

Isolation vs Shutdown

Action	Description
Isolation	System remains active but without network connection
Shutdown	System is completely shut down

In many cases isolation is preferred, since important forensic information in memory can be preserved.

8.4 Containment Examples

Malware Infection

- isolate the system
- perform EDR analysis
- remove malware

Compromised Account

- disable account
- reset password

- perform log analysis

Ransomware

- isolate infected systems
- verify backup integrity
- prevent further spread

9. Recovery / Восстановление

9 Recovery

The recovery phase begins after the attack has been successfully contained.

The goal of this phase is to restore affected systems and safely resume normal operations.

Definition:

Recovery describes the restoration of affected systems and data after a security incident.

Objectives of the Recovery Phase

- restoration of compromised systems
- ensure system integrity
- prevent renewed attacks
- return to normal operations

Recovery Strategies

- reinstall the system
- restore from backup
- clean the system
- correct configuration

Recovery Workflow

System Analysis → System Restoration →

Return to Operation

9.1 Recovery Best Practices

Improper recovery can lead to an attacker gaining access to systems again.

Therefore certain best practices should be followed during the recovery phase.

Integrity Verification

Before a system is put back into operation, it must be verified that it has been completely cleaned.

- perform malware scans
- verify system files
- analyze configurations

System Monitoring

After recovery, systems should be monitored more intensively to detect potential renewed attacks.

- enhanced log analysis
- EDR monitoring
- network monitoring

Ransomware Recovery Example

Typical Recovery Process:

- isolate infected systems
- reinstall systems
- restore backups

- reset passwords
- activate monitoring

Take-away

Recovery does not only mean bringing systems back online, but ensuring that the attack has been completely removed.

10. Post-Incident / После инцидента

10 Post-Incident Activities

After a security incident has been successfully handled, the post-incident phase begins.

In this phase, the incident is analyzed in order to learn from it and improve future security measures.

Definition:

Post-incident activities include all measures that are carried out after a security incident has been resolved.

Objectives of the Post-Incident Phase

- analysis of the security incident
- identification of the cause
- improvement of security measures
- optimization of the incident response process

Lessons Learned Meeting

Many organizations conduct what is known as a lessons learned meeting after an incident.

In this meeting, the incident response team analyzes the entire incident.

Typical Questions:

- How was the attack detected?
- Which systems were affected?
- Which measures were successful?
- Which improvements are necessary?

Incident Documentation

Every security incident should be documented comprehensively. This documentation serves as the basis for future security analyses.

- description of the incident
- timeline
- affected systems
- measures taken
- recommended improvements

10.1 Measuring IRP Effectiveness

To evaluate the effectiveness of an incident response plan, many organizations use various metrics.

These metrics help improve the ability to respond to security incidents.

Important Incident Response Metrics

Metric	Description
MTTD	Mean Time to Detect
MTTC	Mean Time to Contain
MTTR	Mean Time to Recover

Why Metrics Are Important

These metrics help organizations analyze and improve their incident response processes.

Reducing these times can significantly reduce the impact of security incidents.

Take-away

Organizations should regularly analyze and optimize incident response processes in order to respond more quickly and efficiently to security

incidents.

11. Group Exercise / Групповое упражнение

11 Group Exercise – Incident Response Planning

In this group exercise, participants take on the role of leaders in the military IT security domain.

The objective of the exercise is to develop a structured incident response plan for various security incidents.

Role:

Participants act as leaders in the military IT-Security domain.

Task

For each incident scenario, participants must:

- define detection mechanisms
- assess and classify the incident and accompanying risks
- define roles and responsibilities including communication channels
- define containment and recovery processes (flow diagram)
- plan post-incident actions

Incident 1 – Water Intrusion in a Server Facility

Details

- Flooded server room at headquarters
- Cooling system failure causes water leakage

Hints

- hardware destruction
- short circuits
- system downtime

Incident 2 – Unauthorized Access / Physical Security Breach

Details

- Administrator reports two broken door locks
- Administrator did not check further until now

Hints

- rogue hardware
- stolen devices
- system intrusion

Incident 3 – External DDoS Attack

Details

- Massive traffic flood targeting headquarters
- Headquarters provides critical systems to other command posts and units

Hints

- communication breakdown
- cascading system failures
- possible diversion for other attacks

Incident 4 – Internal System Compromise

Details

- suspicious actions detected inside a system
- compromised administrator account

Hints

- data manipulation
- cascading compromise

- privilege escalation